

Xiangling Xu (许湘灵)

My Website

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G Google Scholar

ID Orcid



Research Interests: Quantum information, quantum foundations, operator algebras, noncommutative polynomials optimization

Education

- 2023 – now **Ph.D., Inria Saclay Île-de-France & École doctorale IP Paris**
Supervisor: Marc-Olivier Renou
Project: *Quantum correlations in causal structures: characterization and application*
- 2020 – 2023 **M.Sc. Mathematics, ETH Zürich**
Master Thesis: *Quantum Nonlocality in Bilocal Networks: An Operator Algebraic Perspective*
Semester Project: *Quantum BCOV Theory on Calabi-Yau Manifold*
CGPA: 5.69/6.00
- 2016 – 2020 **B.Sc. Mathematics and Physics, University of Toronto**
Projects: *Noncommutative Geometry in Gravity, Large-Scale Cosmological Structures, Optical Simulation, Biomechanical Simulation*
CGPA: 3.95/4.00

Publications and Preprints

- 1 M. Baroni, I. Klep, D. Leichtle, M.-O. Renou, I. Šupić, L. Tendick, and **X. Xu**, *Quantitative quantum soundness for all multipartite compiled nonlocal games*, 2025. arXiv: 2509.25145 [quant-ph]. [URL: https://arxiv.org/abs/2509.25145](https://arxiv.org/abs/2509.25145).
- 2 I. Klep, C. Paddock, M.-O. Renou, S. Schmidt, L. Tendick, **X. Xu**, and Y. Zhao, *Quantitative quantum soundness for bipartite compiled bell games via the sequential npa hierarchy*, 2025. arXiv: 2507.17006 [quant-ph]. [URL: https://arxiv.org/abs/2507.17006](https://arxiv.org/abs/2507.17006).
- 3 **X. Xu**, M.-O. Renou, and I. Klep, *Quantitative tsirelson's theorems via approximate schur's lemma and probabilistic stampfli's theorems*, 2025. arXiv: 2505.22309 [quant-ph]. [URL: https://arxiv.org/abs/2505.22309](https://arxiv.org/abs/2505.22309).
- 4 M.-O. Renou, **X. Xu**, and L. T. Ligthart, *Two convergent npa-like hierarchies for the quantum bilocal scenario*, 2024. arXiv: 2210.09065 [quant-ph]. [URL: https://arxiv.org/abs/2210.09065](https://arxiv.org/abs/2210.09065).
- 5 X. Luo, **X. Xu**, and X. Wang, "On the kinematic morphology around haloes," *Monthly Notices of the Royal Astronomical Society*, vol. 518, no. 4, pp. 6059–6064, 2023. [DOI: https://doi.org/10.1093/mnras/stac3500](https://doi.org/10.1093/mnras/stac3500).
- 6 **X. Xu**, "Quantum nonlocality in bilocal networks: An operator algebraic perspective," M.S. thesis, ETH Zurich, 2023. [DOI: https://doi.org/10.3929/ethz-b-000613670](https://doi.org/10.3929/ethz-b-000613670).

Conferences, Seminars, Workshops

Contributed Talks

Oct 2025 **YQIS25**, Institute of Photonic Sciences (ICFO)

Conferences, Seminars, Workshops (continued)

- Jul 2025  *IWOTA 2025 Twente*, University of Twente
- Jun 2025  *IQC-PCQT-Quantum Saclay Workshop*, Sorbonne University
- Sep 2024  *Causalworlds*, Perimeter Institute for Theoretical Physics
- Nov 2023  *Workshop Defi EQIP*, Inria Lyon
- Oct 2023  *Saclay Quantum Seminar*, Inria Saclay

Poster Sessions

- Nov 2024  *Colloquium GDR TeQ*, Sorbonne University
-  *YQIS24*, Inria de Paris

Other Participation

- Mar 2025  828. WE-Heraeus-Seminar: Operator Theory and Polynomial Optimization in Quantum Information Theory, Bad Honnef
- Sep 2024  VQF Vienna Quantum Foundations Conference, IQOQI Vienna
- Nov 2023  POP23 Workshop on Future Trends in Polynomial Optimization, LAAS-CNRS Toulouse

Collaborations

Group Visits

- Sep 2024  *Igor Klep*, University of Ljubljana
Two-week visit funded by Erasmus+ Mobility program
- Jul 2024  *William Slofstra*, Institute for Quantum Computing, University of Waterloo
Two-week visit
- Dec 2023  *David Gross*, University of Cologne
One-week visit

Organizations

- Mar 2025 - Now  Coorganizer for *Seminars in Commutative and Noncommutative Polynomial Optimization for Quantum Theory*
Coordinating the Gaspard Monge Visiting Professor program for Igor Klep
Co-planning the seminar and the schedule for invited speakers
- Mar 2025  Coorganizer for 828. *WE-Heraeus-Seminar at Bad Honnef*
Exploring Operator Theory and Polynomial Optimization in Quantum Information Theory with world-renowned experts
Coordinating with the organizers and inviting some of the speakers
- Feb 2024 - Now  Coorganizer of *Project COMPUTE for Quanterra*
Coordinating with the members
Making the project poster and website
- Sep 2023  Coorganizer for *Saclay Quantum Seminar on NPA hierarchy*
Planning and organizing the one-day program to bring all quantum groups at Saclay together

Awards and Scholarships

- 2024  *Erasmus+ Mobility Program*, funds for visiting the University of Ljubljana

Awards and Scholarships (continued)

2020	■	<i>McNab Undergraduate Scholarship</i> , University of Toronto
2017 - 2020	■	<i>Dean's List Scholar</i> , University of Toronto
2018	■	<i>The Class of 3To and Associates Scholarship in Mathematics and Physics</i> , University of Toronto
	■	<i>International Exchange Award</i> , University of Toronto
	■	<i>Bronze Medal</i> , 2017 University Physics Competition
2017	■	<i>Eileen Doman Award</i> , University of Toronto
	■	<i>Summer Undergraduate Research Program</i> , Canadian Institute for Theoretical Astrophysics
2016	■	<i>President's Scholar of Excellence Program</i> , University of Toronto

Teaching Experience

Lecture

Nov 2024	■	<i>Recent trends in parallel, distributed, and quantum computing</i> Master 2 QDCS, Université Paris-Saclay
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Teaching Assistance

2022-2023	■	<i>Linear Algebra I & II</i> Department of Mathematics, ETH Zürich
2021-2022	■	<i>Analysis I & II</i> Department of Mathematics, ETH Zürich

Skills

Languages	■	Chinese (native), English (fluent), German (beginner)
Coding	■	Python (Ncpolzdpa), Matlab (Yalmip), Mathematica, C Language, Hugo on GitHub, L ^A T _E X

Referees

Available on Request